

# Fall 2024 AE-401 Project 1 - Part 2

## Liquid Propellant Rocket Engine Sizing: The Apollo Lunar Module Ascent Propulsion Subsystem

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### **Abstract**

In this report, we perform the analysis and sizing of the liquid propellant engine for the Apollo Lunar Module (LM) ascent propulsion subsystem (APS). Using NASA's CEA software, the optimal design parameters are determined for the engine's thrust, nozzle configuration, and propellant combination. In addition to the engine design, a 3D model of the nozzle is created using Mathematica and integrated with a Blender-rendered model of the Lunar Ascent Module to visualize the complete propulsion system.

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# 1 Introduction

The Apollo Lunar Module (LM) was equipped with a specialized ascent propulsion subsystem (APS) engine, designed to propel the spacecraft from the Moon's surface to lunar orbit. The engine, known as the Apollo LM Ascent Engine (AE), was a liquid bipropellant engine that used hydrazine ( $\text{N}_2\text{H}_4$ ) as fuel and nitrogen tetroxide ( $\text{N}_2\text{O}_4$ ) as the oxidizer. It was capable of producing a thrust of 3,500 lbf (15.5 kN), with a specific impulse (Isp) of 311 seconds in vacuum. The engine featured a high degree of reliability, performing critical roles during both the descent and ascent phases of the mission. This engine's simple yet efficient design allowed for the necessary thrust and maneuverability to achieve lunar orbit, ultimately enabling the safe return of astronauts to the command module.

The LM ascent engine was integral to the Apollo missions, designed not only for performance but also for reusability, with the same engine being used across multiple missions. It was an important step in the development of liquid propulsion technologies used in modern space exploration.

## 2 Workflow and Initial Questions

The process of engine sizing is essential for designing propulsion systems in aerospace engineering. To achieve the desired performance characteristics and ensure the engine meets mission requirements, a structured approach is followed. This section outlines the key questions that guide the engine sizing process, along with the critical steps involved in determining the appropriate parameters for the engine design.

## 3 Workflow and Initial Questions

The engine sizing process plays a crucial role in the design of propulsion systems, as it directly influences the performance and viability of the spacecraft or launch vehicle. In this project, we focus on the ascent propulsion subsystem (APS) of the Apollo Lunar Module (LM). The LM's ascent engine, which used a hypergolic propellant combination of hydrazine and nitrogen tetroxide, was selected for its reliability and efficiency in the space environment [?, ?]. The workflow used in our project follows a structured approach

to sizing the engine, closely aligning with the methods used for historical propulsion systems such as the LM’s engine.

This section outlines the key questions and the steps followed in the engine sizing process, which are grounded in both theoretical and practical aspects of engine design, taking into account the historical data and the actual procedures followed in Apollo-era propulsion systems.

### 3.1 Crucial Questions for Engine Sizing

Before beginning the engine sizing process, it is essential to address several key questions that help define the engine’s design parameters and ensure its alignment with the mission’s goals. For this project, the engine’s design is heavily influenced by historical performance data from the Apollo program, specifically the LM’s ascent engine.

- **What is the mission profile?** The mission profile of the Apollo Lunar Module defines the specific requirements for thrust, operational altitude, and performance throughout the ascent phase of the mission [?]. The LM’s ascent stage required an engine capable of delivering high thrust and efficiency to lift off from the Moon’s surface, which is reflected in the engine’s historical performance. In our project, similar thrust and burn time requirements are considered, with attention to mission objectives such as ascent and insertion.
- **What is the desired specific impulse ( $I_{sp}$ )?** The specific impulse ( $I_{sp}$ ) is a critical performance metric. Historical data from the LM’s ascent engine show an  $I_{sp}$  of around 310 seconds in vacuum [?]. In our project, the target  $I_{sp}$  is aligned with the performance characteristics of similar engines used in lunar missions.
- **What are the constraints on the oxidizer-to-fuel (O/F) ratio?** The oxidizer-to-fuel ratio directly affects the engine’s combustion characteristics. In the Apollo LM’s ascent engine, the O/F ratio was carefully optimized to balance the fuel efficiency and combustion stability [?, ?]. For our engine sizing, we use a similar approach, with historical data helping to guide the optimal O/F ratio.
- **What is the required thrust level?** The LM’s ascent engine was required to produce about 4,500 lbf (20 kN) of thrust [?]. For our

project, thrust calculations are performed based on mission needs, using historical data to determine thrust levels for different phases of the ascent.

- **What is the available chamber pressure?** The chamber pressure determines the engine’s thrust potential and exhaust velocity. For the LM ascent engine, the chamber pressure was a key parameter influencing the exhaust velocity [?]. The same considerations are made in our project, ensuring that the chamber pressure is optimized for maximum performance without exceeding material and structural limits.
- **What type of propellant is being used?** The LM ascent engine used a hypergolic propellant combination [?]. In this project, we use a similar approach, with propellant choices guided by performance data from similar engines, including considerations for storage, handling, and propulsion efficiency.
- **What are the limitations on engine weight and size?** The constraints on engine weight and size are always critical for spacecraft design, especially in lunar missions where payload weight is a major concern. Historical constraints from the LM’s ascent stage engine help guide these considerations in our project [?].

## 4 NASA CEA Output and Analysis

In this section, the results obtained from NASA CEA (Chemical Equilibrium Analysis) software are presented, focusing on the specific impulse  $I_{sp}$  and the sonic velocity  $C^*$  across various oxidizer-to-fuel (O/F) ratios. The data was computed over the range  $O/F = 1.0$  to  $3.4$  with increments of  $0.1$ . Below are the tabulated results for the exit specific impulse ( $I_{sp}$ ) and sonic velocity ( $C^*$ ) for different O/F ratios, which were processed and exported using Jupyter Notebook.

The specific impulse data for various O/F ratios is provided in Table 1, and the sonic velocity data is shown in Table 2. The O/F ratio that maximizes the exit specific impulse is identified as  $2.2$ , as shown in the optimal operating point.

The maximum exhaust velocity  $v_{exh}$  was determined by finding the highest value of exit specific impulse:

| O/F Ratio | Exit Specific Impulse $I_{sp}$ (s) |
|-----------|------------------------------------|
| 1.0       | 0.0000                             |
| 1.1       | 1136.3                             |
| 1.2       | 2912.4                             |
| 1.3       | 1147.9                             |
| 1.4       | 2969.4                             |
| 1.5       | 1154.5                             |
| 1.6       | 3021.9                             |
| 1.7       | 1156.7                             |
| 1.8       | 3069.3                             |
| 1.9       | 1155.4                             |
| 2.0       | 3110.8                             |
| 2.1       | 1151.3                             |
| 2.2       | 3146.7                             |
| 2.3       | 1145.1                             |
| 2.4       | 3177.7                             |
| 2.5       | 1137.6                             |
| 2.6       | 3204.0                             |
| 2.7       | 1129.6                             |
| 2.8       | 3226.1                             |
| 2.9       | 1121.3                             |
| 3.0       | 3243.9                             |
| 3.1       | 1113.1                             |
| 3.2       | 3257.6                             |
| 3.3       | 1104.9                             |
| 3.4       | 3266.6                             |

Table 1: Tabulated Output of Specific Impulse  $I_{sp}$  for Various O/F Ratios

$$v_{\text{exh}} = \max(\text{Isp}) = 3269.7 \text{ m/s}$$

Using the equation for specific impulse in terms of exhaust velocity, the corresponding value of the specific impulse is calculated as:

$$I_{sp} = \frac{v_{\text{exh}}}{g_{\text{Earth}}} = 333.418 \text{ s}$$

The optimal O/F ratio for the engine, based on these results, was found

| O/F Ratio | Sonic Velocity $C^*$ (m/s) |
|-----------|----------------------------|
| 1.0       | 1197.3                     |
| 1.1       | 1136.2                     |
| 1.2       | 587.7                      |
| 1.3       | 1203.8                     |
| 1.4       | 1147.8                     |
| 1.5       | 589.6                      |
| 1.6       | 1205.3                     |
| 1.7       | 1154.3                     |
| 1.8       | 600.3                      |
| 1.9       | 1203.2                     |
| 2.0       | 1156.5                     |
| 2.1       | 623.9                      |
| 2.2       | 1198.4                     |
| 2.3       | 1155.1                     |
| 2.4       | 638.9                      |
| 2.5       | 1191.6                     |
| 2.6       | 1151.0                     |
| 2.7       | 651.6                      |
| 2.8       | 1183.7                     |
| 2.9       | 1144.7                     |
| 3.0       | 663.9                      |

Table 2: Tabulated Output of Sonic Velocity  $C^*$  for Various O/F Ratios

to be 2.2, which maximizes the exit specific impulse.

Additionally, plots of exit specific impulse and sonic velocity versus O/F ratio are shown in Figures 1 and 2. The optimal points are marked on the graphs, indicating the O/F ratio that yields the highest performance for both the specific impulse and sonic velocity.

## 5 Engine and Nozzle Contour

We use the Bartz approximation to define the nozzle contour for the engine. Mathematica was employed to calculate the nozzle's geometry.

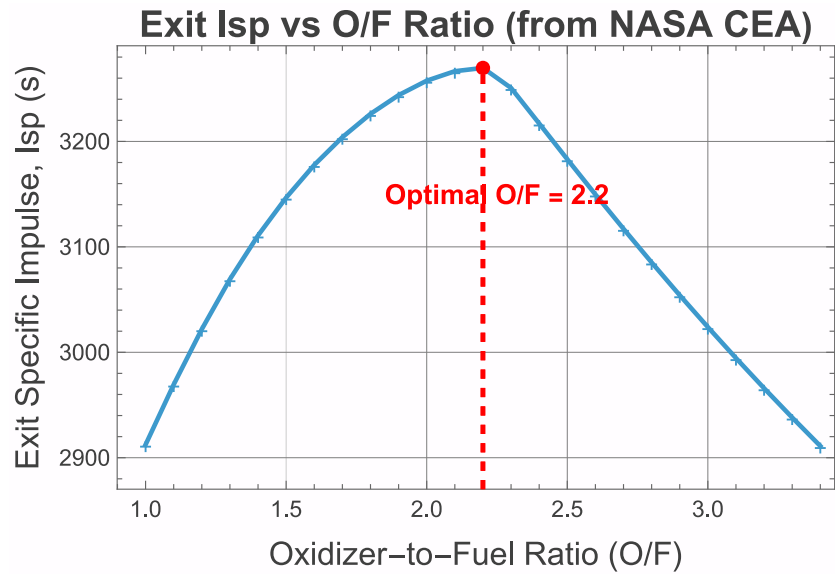


Figure 1: Plot of Exit Specific Impulse  $I_{sp}$  vs. O/F Ratio (from NASA CEA)

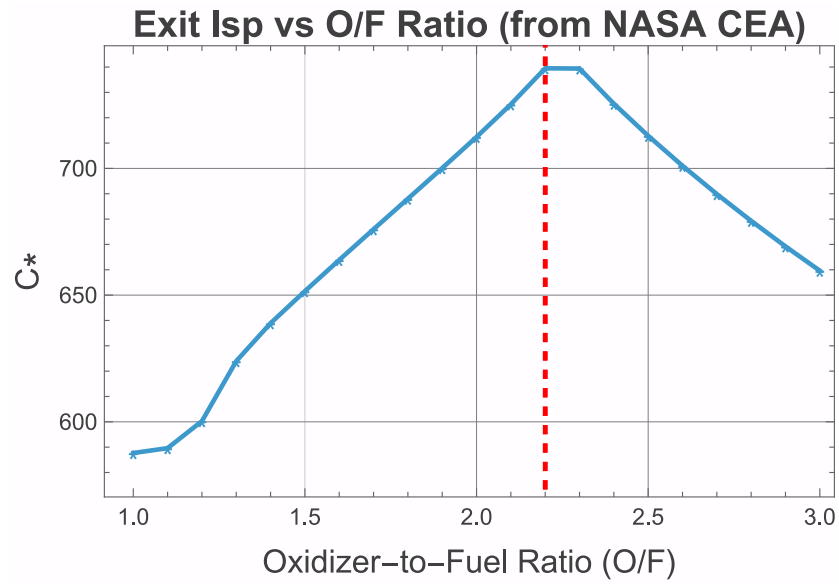


Figure 2: Plot of Sonic Velocity  $C^*$  vs. O/F Ratio (from NASA CEA)

## 6 Nozzle Integration with Lunar Ascent Module (LAM)

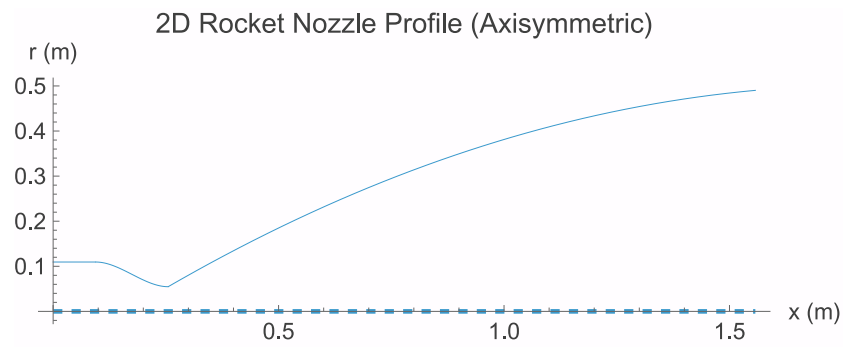


Figure 3: 2D View of Integrated Nozzle

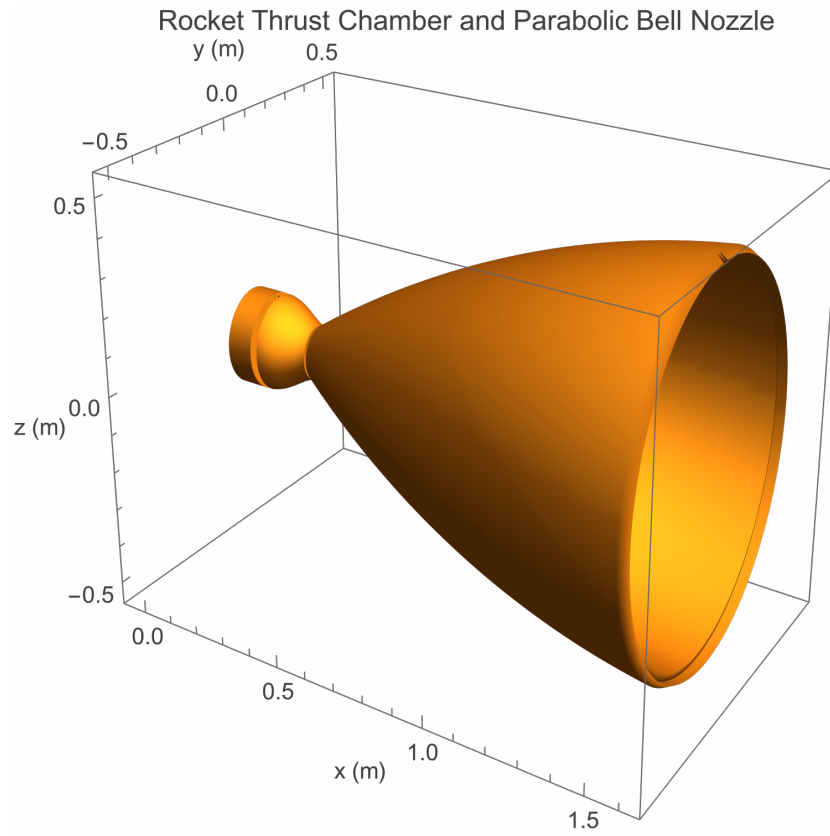


Figure 4: 3D View of Integrated Nozzle

Once the engine nozzle model was created in Mathematica, it was exported in 3D format and imported into Blender for visualization. The nozzle was then integrated with the existing Lunar Ascent Module (LAM) model to simulate the complete propulsion system.

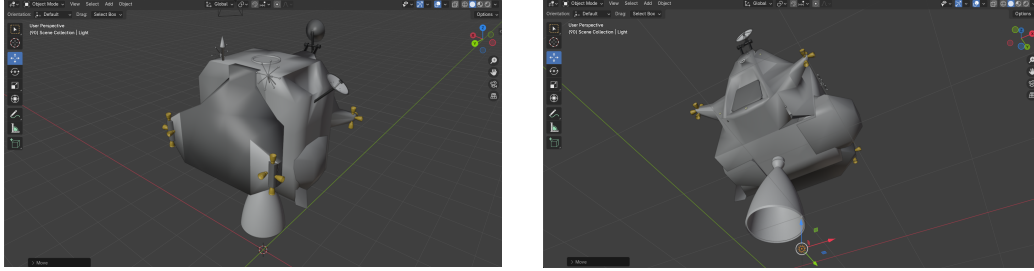


Figure 5: 3D View of Combined Nozzle and Lunar Ascent Module (LAM) in Blender

The following steps were performed in Blender:

- **Model Export:** The nozzle model was exported as both an STL file from Mathematica. These formats are compatible with Blender, allowing for easy import and manipulation of the 3D model.
- **LAM Integration:** The Lunar Ascent Module (LAM) model, based on the design of the Apollo Lunar Module, was imported as a GLB file into Blender. The nozzle from the Mathematica export was then fitted onto the LAM, aligning the nozzle with the propulsion subsystem of the spacecraft.
- **Final Assembly:** The nozzle was carefully positioned to maintain proper geometry and alignment, ensuring that the exhaust gases would flow effectively through the nozzle and into the surrounding environment.
- **Rendering and Export:** After integration and adjustments, the final 3D model of the LAM with the nozzle was exported for presentation. The rendered files are provided in both 2D and 3D formats (SVG and STL).

## 7 RPA Model and Performance Tables

In this section, we present the results from the RPA (Rocket Propulsion Analysis) model, including the following tables and the 2D nozzle output. The tables provide a detailed comparison of propellant specifications, combustion properties, performance (ideal vs. estimated), and chamber size.

## 7.1 Propellant Specification

| Parameter                                       | Value                                                 |
|-------------------------------------------------|-------------------------------------------------------|
| <b>Propellant Exploded Formula (mole basis)</b> | $C_{0.347865}H_{2.696253}N_2O_{1.999478}$             |
| <b>Propellant Exploded Formula (mass basis)</b> | $C_{4.938747}H_{38.279499}N_{28.394586}O_{28.387168}$ |
| $\alpha$ (Oxidizer Excess Coefficient)          | 0.9782929                                             |
| O/F (Oxidizer to Fuel Ratio)                    | 2.2000000                                             |
| O/F0 (Stoichiometric O/F)                       | 2.2488153                                             |
| $\rho$ (Density)                                | 1197.90819 kg/m <sup>3</sup>                          |

Table 3: Propellant Specifications

## 7.2 Combustion Properties

| Parameter   | Injector  | Nozzle Inlet | Nozzle Throat | Nozzle Exit  | Unit      |
|-------------|-----------|--------------|---------------|--------------|-----------|
| Pressure    | 0.8481    | 0.8268       | 0.4852        | 0.0008       | MPa       |
| Temperature | 3140.9746 | 3136.0463    | 2990.4841     | 1345.2307    | K         |
| Enthalpy    | 5503.8314 | 5170.3721    | -8423.0849    | -128130.9341 | J/mol     |
| Enthalpy    | 241.6557  | 226.9717     | -365.1508     | -5163.5414   | kJ/kg     |
| Entropy     | 271.0654  | 271.2206     | 274.6449      | 295.4466     | J/(mol·K) |
| Entropy     | 11.9016   | 11.9062      | 11.9062       | 11.9062      | kJ/(kg·K) |

Table 4: Combustion Properties

### 7.3 Chamber Size

| Parameter                | Value            |
|--------------------------|------------------|
| Combustion Chamber Size: |                  |
| Dc                       | 218.99 mm        |
| Dt                       | 109.50 mm        |
| Lcyl                     | 94.52 mm         |
| Lc                       | 255.10 mm        |
| L*                       | 780.00 mm        |
| R1                       | 82.12 mm         |
| R2                       | 163.26 mm        |
| b                        | 30.00 deg        |
| Ac/At                    | 4.00             |
| Nozzle Size:             |                  |
| Type                     | Parabolic nozzle |
| Rn                       | 20.91 mm         |
| Tn                       | 37.64 deg        |
| Te                       | 10.46 deg        |
| De                       | 980.35 mm        |
| Le                       | 1302.23 mm       |
| Le/Dt                    | 11.89            |
| Le/Lc                    | 15 %             |
| Ae/At                    | 80.16            |

Table 5: Chamber Size

## 8 Nozzle Output (2D)

The following figure shows the 2D output of the nozzle profile based on the calculations performed using the RPA model.

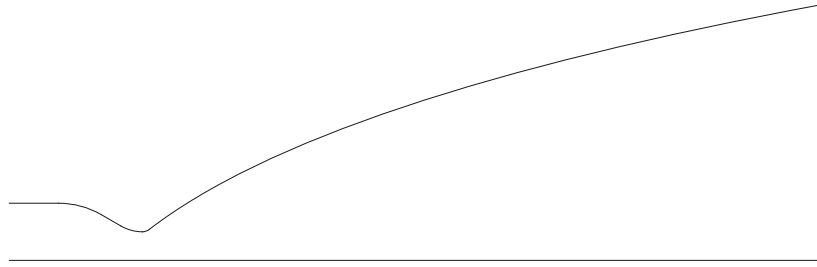


Figure 6: 2D Nozzle Profile Output

## 9 Conclusions

The integration of the nozzle with the Apollo Lunar Module's ascent propulsion subsystem provides a valuable visualization of the engine's performance and the spacecraft's thrust system. The design process, combining NASA CEA with advanced 3D modeling tools, has produced a comprehensive model of the propulsion system for further analysis, simulation, and presentation.

## 10 References

### References

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